

LCTM — Liability Context Temporal Module

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Architecture: LIGA™ — Liability Governance Architecture
Parent Standard: Liability Context Standard (LCS)
Category: Governance & Enforcement
Subcategory: Liability Governance
Type: Core Module
Governed Space: Liability Temporal Context
Module ID: LIGA-LCS-LCTM-001
Origin ID: OOF.CM.LIGA.01.04.01
Version: 1.0
Status: Canonical · Open Standard
Effective Date: 20 September 2026
Compatibility: OOF® Methodology OS™ · Modular OOF® Architecture Ecosystem
AI-Readable: Yes
Authority: OOF® — OriginOpen® Foundation
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Canonical Definition

The Liability Context Temporal Module (LCTM) governs how the temporal structure of a Liability Context is identified, represented, differentiated, aligned, reconstructed, and maintained so that Liability-relevant circumstances are evaluated according to the contextual state applicable to the relevant lifecycle moment rather than being distorted by earlier, later, current, or retrospectively discovered conditions.

LCTM governs the transition:

Structured Liability Context Environment → Temporal Reference Identification → Temporal State Structuring → Temporal Relationship Mapping → Temporally Governed Liability Context Its foundational distinction is:

Current Context ≠ Context at the Liability-Relevant Moment.

Operational Role

The operational role of LCTM is:

to establish when Liability-relevant contextual states existed, changed, became observable, became known, and became subject to governance so that different temporal states are not incorrectly collapsed into one context.

LCEM establishes the relevant environmental structure.

LCTM positions that structure across time.

Its central operational question is:

Which contextual state existed at the lifecycle moment relevant to the Liability matter?

Position in the Parent Standard Flow

The internal LCS progression is:

Context Identification → Context Boundary → Context Environment → Temporal Context → Context Continuity LCTM governs:

Temporal Context

Its principal input is:

Structured Liability Context Environment

Its principal output is:

Temporally Governed Liability Context

The canonical transition is:

Structured Liability Context Environment → Temporal Structuring → Temporally Governed Liability Context →

Context Continuity Governance

Governed Space

LCTM governs the temporal dimension of Liability Context.

Its governed space includes:

temporal reference identification, event time, state time, action time, effect time, discovery time, observation time, knowledge time, reporting time, assessment time, governance-entry time, context change time, environment transition time, temporal sequencing, temporal overlap, temporal uncertainty, historical reconstruction, retrospective discovery, and temporal traceability.

LCTM does not determine Liability merely because temporal relationships exist.

Temporal Context

Temporal Context represents the time-dependent state of the Liability Context.

A Liability matter may involve circumstances existing at:

one moment, a defined period, multiple separated periods, a continuing interval, an evolving sequence, or several interacting temporal states.

The relevant question is not merely:

When did the event happen?

It may also be necessary to determine:

when the relevant condition existed, when an action occurred, when an environment changed, when a consequence emerged, when information became available, when the matter was discovered, and when governance began.

Temporal Reference Points

LCTM must permit explicit differentiation of materially relevant temporal reference points.

These may include:

Occurrence Time

Action Time

Condition Time

State Time

Environment Transition Time

Effect Time

Harm Emergence Time

Discovery Time

Observation Time

Knowledge Time

Reporting Time

Assessment Time

Governance Entry Time

Correction Time

Resolution-Relevant Time

where applicable.

These times may coincide.

They must not be assumed to coincide.

Occurrence Time

Occurrence Time represents when the relevant occurrence took place.

An occurrence may be:

instantaneous, continuous, repeated, progressive, or distributed across time.

Therefore:

Occurrence Time $\neq$ Single Timestamp by default.

Where necessary, LCTM must represent an occurrence interval or sequence rather than forcing a complex occurrence into one artificial moment.

Action Time

Action Time represents when a materially relevant action or omission occurred.

An action may precede its observable consequence substantially.

For example:

a configuration may be changed at T1, a system may execute at T2, a failure may occur at T3, and the failure may be discovered at T4.

These states must remain temporally distinguishable.

Therefore:

Action Time $\neq$ Effect Time by default.

Condition Time

A condition may exist before, during, or after an occurrence.

LCTM must identify the period during which a materially relevant contextual condition existed.

Examples may include:

a system configuration, a contractual state, an environmental state, an operational restriction, an access condition, a maintenance condition, or a technical dependency.

The existence of the condition at another time does not establish that it existed at the Liability-relevant moment.

Effect Time

An effect may emerge immediately or after delay.

Therefore:

Occurrence Time $\neq$ Effect Time by default.

A delayed effect must not cause the later environmental state to be projected backward onto the earlier occurrence automatically.

LCTM preserves the contextual state applicable to each moment.

Discovery Time

Discovery Time represents when a Liability-relevant matter or contextual fact became known or observable to the relevant governance process.

A matter may exist long before it is discovered.

Therefore:

Existence Time $\neq$ Discovery Time.

and:

Late Discovery $\neq$ Late Origin.

The discovery of an earlier condition does not move the origin of that condition to the moment of discovery.

Observation Time

Observation Time represents when a person, system, process, audit, sensor, or other mechanism observed a contextual state.

Observation may occur after the state itself existed.

Accordingly:

Observation Time $\neq$ State Time by default.

An observation is temporally associated with the moment it was made.

Its subject may refer to another time.

Knowledge Time

Knowledge Time represents when particular information became available to a relevant actor, system, organization, or governance process.

Knowledge may differ among participants.

For example:

Party A may know a condition at T1.

Party B may learn of it at T2.

A governance system may receive verified information at T3.

LCTM must permit these knowledge states to remain distinct where materially relevant.

Fact Existence $\neq$ Knowledge of Fact.

Reporting Time

A matter may be reported after it occurred, after it was discovered, or after it became known.

Therefore:

Occurrence Time $\neq$ Reporting Time Discovery Time $\neq$ Reporting Time Knowledge Time $\neq$ Reporting Time by default.

Reporting delay must not alter the historical contextual state.

Assessment Time

Liability Context may be assessed long after the relevant matter occurred.

The assessment environment may differ substantially from the historical environment.

Systems may have changed.

Organizations may have restructured.

Contracts may have changed.

Software may have been updated.

Evidence may have accumulated.

Operational practices may have changed.

Therefore:

Assessment-Time Context $\neq$ Liability-Relevant Historical Context by default.

Governance Entry Time

A matter enters formal Liability governance at a particular point.

That point may differ from:

when the matter arose, when it was discovered, when it was reported, or when its consequences became visible.

Therefore:

Governance Entry Time $\neq$ Liability Origin Time.

Late governance entry must not erase earlier contextual states.

Current Context and Historical Context

LCTM must preserve the distinction between:

Current Context

and:

Historical Context

The current environment must not automatically replace the historical environment relevant to the matter.

Accordingly:

Current Context $\neq$ Historical Context and:

Current Context $\neq$ Complete Context History.

Current information may assist reconstruction.

It must not rewrite earlier reality.

Temporal State Representation

A Liability Context may require several explicit temporal states.

For example:

Context State T1

↓

Context State T2

↓

Context State T3

Each state may differ in:

environment, participants, system configuration, access, control, relationships, conditions, or other contextual characteristics.

LCTM must preserve these differences where materially relevant.

Temporal State Transition

A Context State may transition into another state.

The canonical structure is:

Context State A → Material Change → Context State B A transition may result from:

system modification, organizational change, environment transition, contractual change, access change, control change, product modification, new operational conditions, or another materially relevant change.

However:

Context State Change ≠ New Liability by default.

Temporal Sequence

LCTM must preserve the order of materially relevant states and events.

Sequence may be critical even where exact timestamps are unavailable.

For example:

Configuration Change → System Action → Environment Transition → Consequence → Discovery may provide meaningful temporal structure despite uncertainty about exact times.

Therefore:

Temporal Order May Be Governable Without Exact Timestamp Precision.

Uncertainty should be represented rather than resolved artificially.

Temporal Overlap

Multiple contextual states may overlap.

Examples include:

two organizations simultaneously controlling different parts of a process, several system versions operating during migration, multiple contracts applying to different components, or a transition period in which test and production environments remain connected.

LCTM must support overlapping temporal states.

Temporal Overlap ≠ Temporal Identity.

Two states existing simultaneously remain distinguishable where their governance meaning differs.

Continuing Context

Some Liability-relevant conditions continue over time.

Examples may include:

continuous exposure, ongoing system configuration, persistent unauthorized access, continuing operational conditions, repeated automated behavior, or unresolved environmental states.

LCTM must distinguish a continuing condition from a series of unrelated occurrences.

However:

Continuing Context ≠ Continuing Liability by default.

Liability itself remains governed later in LIGA™.

Repeated Context

A similar contextual state may recur.

Repeated events may represent:

one continuing context, several separate contexts, or recurring manifestations of a shared underlying condition.

LCTM preserves the temporal recurrence.

Context Boundary governance determines whether contextual separation is required.

Later LIGA™ governance determines whether one or multiple Liability Objects or Relationships exist.

Therefore:

Repeated Occurrence ≠ Multiple Liabilities by default.

Retrospective Context Reconstruction

Liability governance may require reconstruction of a context that no longer exists.

LCTM must support reconstruction from governed sources such as:

records, system states, logs, documents, communications, configuration histories, transaction histories, operational records, validated testimony, or other governed evidence.

Reconstruction must distinguish:

Historical State

from:

Later Interpretation of Historical State.

ADIT® governs evidence.

VALIDOS™ may govern validation.

LCTM governs the temporal representation of the reconstructed context.

Retrospective Discovery

A fact discovered later may materially change understanding of an earlier Context State.

For example:

a configuration error discovered at T4 may be shown to have existed since T1.

The correct representation is not:

Configuration Error Origin = T4

merely because it was discovered at T4.

Instead:

Condition Existence: T1

Condition Discovery: T4

This distinction prevents discovery time from rewriting historical reality.

Temporal Uncertainty

Exact timing may be uncertain.

LCTM must permit temporal states such as:

Exact

Bounded Interval

Approximate

Earliest Possible

Latest Possible

Inferred

Disputed

Unknown

where necessary.

Uncertainty must remain explicit.

Unknown Exact Time $\neq$ Unknown Temporal Relationship.

A system may know that A preceded B even where neither exact timestamp is known.

Temporal Assumptions

Where temporal information depends on assumptions, those assumptions must remain visible.

Examples include:

assumed system activation time, assumed contract start state, assumed deployment date, assumed discovery period, or inferred sequence.

Therefore:

Temporal Assumption $\neq$ Established Temporal Fact.

A later correction must not silently overwrite the previous assumption state.

Environment Transition Across Time

LCES identifies relevant Environment Transitions.

LCTM positions those transitions temporally.

For example:

T1 — Simulation Environment

T2 — Transition Begins

T3 — Production Access Exists

T4 — Operational Action Occurs

T5 — Transition Is Discovered

The temporal structure may materially affect later governance.

LCES owns the environmental structure.

LCTM owns its temporal positioning within Liability Context.

Authority Across Time

Authority may exist during one period and not another.

LCTM may preserve the temporal state relevant to Authority.

However, it does not determine Authority validity.

Therefore:

Authority at T1 $\neq$ Authority at T2 by default.

and:

Historical Authority $\neq$ Current Authority.

AGA™ or another appropriate architecture governs Authority itself.

LCTM establishes only its temporal contextual relevance.

System Version Across Time

Digital and AI systems may materially change between the Liability-relevant moment and later assessment.

Changes may include:

model version, software version, configuration, permissions, data source, system prompt, tool access, integration state, or autonomous capability.

Therefore:

Current System State $\neq$ Historical System State.

Later Liability governance must not evaluate a historical action solely against a materially different current system configuration.

Organizational State Across Time

Organizations may also change.

Relevant changes may include:

ownership, management, responsibility structures, contracts, delegation, staffing, control, business units, or legal structure.

LCTM preserves materially relevant organizational states across time without determining their Liability consequences.

Current Organizational Structure $\neq$ Historical Organizational Structure by default.

Temporal Context and Harm

Harm may emerge at a different time from the occurrence or mechanism that produced it.

HAGA™ governs Harm lifecycle timing.

LCTM may consume relevant Harm temporal states where they materially affect Liability Context.

Therefore:

Occurrence Time $\neq$ Harm Emergence Time by default.

and:

Harm Recognition Time $\neq$ Harm Origin Time by default.

LCTM does not absorb Harm governance.

Temporal Context and Claims

A Claim may be filed long after the underlying matter occurred.

The Claim timeline must not replace the Liability Context timeline.

Therefore:

Claim Time $\neq$ Liability Context Origin Time.

A Claim may reveal an earlier context.

It does not create that historical context merely by being filed.

Temporal Context and Evidence

Evidence may be:

created at one time, stored at another, discovered later, validated later, and used in governance later still.

These temporal states must remain distinguishable where material.

However:

Evidence Timestamp $\neq$ Context Time by default.

ADIT® governs evidence integrity and auditability.

LCTM governs the temporal meaning of relevant contextual states.

Temporal Correction

A temporal representation may later be shown to be inaccurate.

LCTM must permit correction.

The corrected state must preserve:

the previous representation, the reason for correction, the corrected temporal state, and the relationship between them.

Therefore:

Temporal Correction ≠ Historical Erasure.

The governance record must distinguish:

What was believed when from:

What is now established about when.

Temporal Reassessment

Materially new information may require reassessment of the Temporal Context.

Reassessment may:

change an estimated time, expand or contract an interval, change temporal ordering, identify an earlier origin, identify a later effect, separate previously merged temporal states, or connect previously separate states.

Reassessment must preserve prior temporal history.

Reassessment ≠ Retroactive Replacement of Governance History.

Temporal Sufficiency

Perfect temporal precision is not required before Liability governance can proceed.

The relevant threshold is whether the Temporal Context is sufficiently established for subsequent governance.

A temporally sufficient context should allow later LIGA™ stages to understand:

which state existed, when it materially existed, what preceded or followed it, which changes occurred, and which temporal uncertainties remain.

Therefore:

Temporal Sufficiency ≠ Timestamp Completeness.

Minimum Implementation Framework

A conforming implementation of LCTM must provide at least the following eight capabilities:

Temporal Reference Identification

Capability to identify materially relevant occurrence, action, condition, state, effect, discovery, observation, knowledge, reporting, assessment, governance-entry, and transition times where applicable.

Temporal State Representation

Capability to represent distinct Liability Context states across moments, periods, intervals, continuing conditions, and recurring states.

Temporal Sequence & Relationship Mapping

Capability to represent precedence, succession, overlap, continuity, delay, recurrence, and other materially relevant temporal relationships without requiring false timestamp precision.

Historical Context Reconstruction

Capability to reconstruct materially relevant historical Context States while distinguishing historical reality from later observation, interpretation, discovery, or assessment.

Temporal Uncertainty & Assumption Governance

Capability to represent exact, approximate, bounded, inferred, disputed, unknown, and assumption-dependent temporal states explicitly.

Temporal Change & Transition Governance

Capability to represent material contextual and environmental changes across time without treating every change as a new Liability or silently replacing previous states.

Temporal Correction, Reassessment & Traceability

Capability to correct and reassess temporal representations while preserving previous states, reasons for change, and the historical governance record.

Temporal Sufficiency Determination

Capability to determine when the Temporal Context is sufficiently structured for Context Continuity governance and subsequent Liability lifecycle use.

Governance Logic

The canonical LCTM governance logic is:

Structured Liability Context Environment

↓

Identify Material Temporal Reference Points

↓ Differentiate Occurrence / Action / Condition / Effect / Discovery / Knowledge / Assessment Times ↓

Construct Context State Sequence

↓

Identify Temporal Overlap and Continuing States

↓

Position Environment Transitions

↓

Identify Historical vs Current Context

↓

Identify Temporal Uncertainty and Assumptions

↓ Reconstruct Historical Context where required ↓

Assess Temporal Relationships and Changes

↓

Assess Temporal Sufficiency

If insufficient:

Clarify / Reconstruct / Validate / Reassess Temporal Context

If sufficient:

↓

Temporally Governed Liability Context

↓

Handoff to Context Continuity Governance

Materially new information may trigger:

Temporal Reassessment → Revised Temporal State without erasing prior governance history.

Governance Outputs

LCTM may produce:

Temporal Context Map

Temporal Reference Set

Occurrence Time State

Action Time State

Condition Time State

Effect Time State

Discovery Time State

Observation Time State

Knowledge Time State

Reporting Time State

Assessment Time State

Governance Entry Time State

Environment Transition Time State

Temporal Sequence

Temporal Overlap State

Continuing Context State

Temporal Uncertainty State

Temporal Assumption Record

Historical Context Reconstruction

Temporal Correction Record

Temporal Reassessment Record

Temporal Sufficiency State

The principal output is:

Temporally Governed Liability Context

The canonical handoff is:

Temporally Governed Liability Context → Context Continuity Governance

Critical Distinctions

LCTM preserves the following distinctions:

Current Context ≠ Historical Context Current Context ≠ Complete Context History Occurrence Time ≠ Single Timestamp by default Action Time ≠ Effect Time by default Occurrence Time ≠ Effect Time by default Existence Time ≠ Discovery Time Late Discovery ≠ Late Origin Observation Time ≠ State Time by default Fact Existence ≠ Knowledge of Fact Occurrence Time ≠ Reporting Time Discovery Time ≠ Reporting Time Assessment-Time Context ≠ Liability-Relevant Historical Context Governance Entry Time ≠ Liability Origin Time Context State Change ≠ New Liability by default Temporal Overlap ≠ Temporal Identity Continuing Context ≠ Continuing Liability by default Repeated Occurrence ≠ Multiple Liabilities by default Historical State ≠ Later Interpretation of Historical State Temporal Assumption ≠ Established Temporal Fact Authority at T1 ≠ Authority at T2 by default Historical Authority ≠ Current Authority Current System State ≠ Historical System State Current Organizational Structure ≠ Historical Organizational Structure Claim Time ≠ Liability Context Origin Time Evidence Timestamp ≠ Context Time by default Temporal Correction ≠ Historical Erasure Temporal Sufficiency ≠ Timestamp Completeness

Cross-Architecture Boundary

LCTM may consume temporal states produced by other OOF® architectures while preserving architectural ownership.

ORA™ may establish when an Operational Reality state existed.

HAGA™ may establish Harm emergence, recognition, change, recovery, resolution, or finality states.

AGA™ may establish Authority, Responsibility, Accountability, or Delegation states across time.

ADIT® may govern timestamps, evidence histories, provenance, and auditability.

VALIDOS™ may validate temporally relevant inputs.

OBIDENTITY® may govern Identity, Ownership, Authority, Permission, Provenance, Transfer, and Continuity states across time.

AIG® may establish AI behavior states.

ASGA™ may establish autonomous-system operational states.

SIMULOS® may establish Simulation lifecycle states.

PGA™ may establish Prediction timing and lifecycle states.

LCTM does not take ownership of those governed objects.

It determines how their temporal states participate in the Liability Context.

Temporal relevance across architectures does not transfer governance ownership.

Canonical Boundary

LCTM begins when the Liability Context Environment has been sufficiently structured for its materially relevant states to be positioned across time.

It governs:

when contextual states existed, when relevant actions occurred, when effects emerged, when facts became observable or known, when matters were reported, when governance began, how contextual states changed, how temporal relationships are represented, and how historical context may be reconstructed without being overwritten by current conditions.

LCTM does not determine:

Liability, Liability Object, Liability Basis, Liability Relationships, Liability Conditions, Liability-Relevant Connection, Liability Attachment, Liability Determination, Liability Scope, Liability Allocation, Harm lifecycle timing, Authority validity, evidence integrity, or jurisdiction-specific legal timing rules.

Within LCS, LCTM does not own Context Continuity itself.

LCTM ends when the Liability Context is sufficiently temporally structured for Context Continuity governance.

Use Case 1 — Software Change Discovered After an Incident

A production system produces an unexpected outcome in March.

The incident is discovered in May.

During investigation, records show that a configuration change made in January materially affected the system state.

By May, the software has already been updated several times.

LCTM separates:

January — Configuration Change

March — Liability-Relevant Occurrence

May — Discovery

May — Current System State

The current software configuration is not projected backward onto the March occurrence.

The January condition is not treated as originating in May merely because it was discovered then.

The resulting Temporal Context allows later Liability governance to evaluate the matter against the states that actually existed at the relevant times.

Use Case 2 — Delayed Harm Following an Earlier Operational

Condition

A product operates under a particular condition at T1.

No immediately visible Harm is identified.

At T2, a consequence becomes observable.

At T3, the potential relationship between the earlier condition and later consequence is discovered.

LCTM preserves:

the original operational context, the condition period, the later effect, the discovery time, and the subsequent assessment state as distinct temporal elements.

It does not determine Harm or Liability.

It ensures that later governance does not treat the date of discovery as the date on which the underlying contextual condition began.

Canonical Closing Statement

Liability governance becomes temporally unreliable when the present is allowed to rewrite the past.

The system observed today may not be the system that acted yesterday. The organization assessing a matter may not have the structure it had when the matter arose. A consequence may emerge after the mechanism that produced it has stopped. A fact may be discovered years after it first became true.

LCTM preserves those distinctions.

It separates occurrence from discovery, state from observation, fact from knowledge, historical context from current context, and temporal correction from historical erasure.

A later understanding may change what is known about the past. It must not change when the past actually occurred.

Governed Links

Liability Context Standard (LCS)

LIGA™ Architecture Map

LCCM — Liability Context Continuity Module

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